Problem 8.7

An investor deposits \$1,000 today. The annual compound rate of discount is 6%. What is the accumulated value of the investment at the end of 10 years?

Solution

Let $d = 0.06$ be the annual effective discount rate. The equivalent effective interest rate i is

$$i = \frac{d}{1-d}.$$

Thus,

$$i = \frac{0.06}{0.94} \approx 0.0638298.$$

Now accumulate \$1,000 for 10 years at this effective interest rate:

$$X = 1000(1+i)^{10} = 1000(1.0638298)^{10}.$$

Therefore,

$$X \approx 1856.6133.$$

So the accumulated value at the end of 10 years is

$$\boxed{\$1856.61}.$$